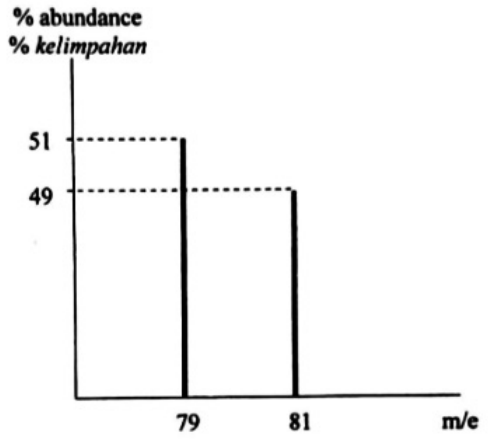
SUGGESTED ANSWER PSPM 1
SESI 2018/2019

1. a) Bromine has proton number of 35. FIGURE below shows a mass spectrum of bromine.



i. Write the notations for all isotopes of bromine.

$^{79}_{35}Br$ **and** $^{81}_{35}Br$

ii. Calculate the relative atomic mass of bromine.

Average atomic mass $= \dfrac{\sum(isotopic\ mass \times abundance)}{\sum abundance}$

$= \dfrac{(79\times 51)+(81\times 49)}{(49+51)}$

= 79.98

∴ relative atomic mass = <u>79.98</u>

b) A reagent bottle contains a stock solution of 0.90% by mass of sodium chloride, NaCl. The density of the solution is 1.00 g cm^{-3}. Calculate

i. The mole fraction of NaCl.

Assume mass of solution = 100 g

Mass of NaCl = 0.90 g

Mole of NaCl = $\dfrac{0.90\ g}{58.5\ g\ mol}$ **= 0.0154 mol**

Mass of H_2O = 100 g – 0.90 g = 99.1 g

Mole of H_2O = $\dfrac{99.1\ g}{18\ g\ mol}$ **= 5.506 mol**

∴Mole fraction of NaCl = $\frac{0.0154\ mol}{(0.0154+5.506)\ mol}$

= **$\underline{2.79\times10^{-3}}$**

ii. The molality of NaCl solution.

Molality = $\frac{\text{Mole of NaCl (mol)}}{\text{mass of solvent (kg)}}$

= $\frac{0.0154\text{ mol}}{0.0991\text{ kg}}$

= **$\underline{0.155\text{ m}}$**

iii. The volume of the stock solution required to prepare 100 mL of 0.01 M NaCl solution.

Density of solution = $\frac{\text{Mass of solution (g)}}{\text{volume of solution (cm}^3\text{)}}$

1.00 g cm^{-3} = $\frac{100\text{ g}}{\text{volume of solution (cm}^3\text{)}}$

Volume of solution = 100 cm^3

Molarity = $\frac{\text{Mole of NaCl (mol)}}{\text{Volume of solution (L)}}$

= $\frac{0.0154\text{ mol}}{0.1\text{ L}}$

= **$\underline{0.154\text{ M}}$**

Mole of diluted NaCl solution = (0.1 L)(0.01 M)

= 1×10^{-3} mol

∴Volume NaCl solution needed = $\frac{1\times10^{-3}\text{ mol}}{0.154\text{ M}}$

= **$\underline{6.49\times10^{-3}\text{ L}}$**

c) Silicon tetrachloride, $SiCl_4$ can be prepared by heating silicon in excess chlorine gas.

$$Si(s)+2Cl_2(g)\rightarrow SiCl_4(l)$$

i. Calculate the mass of silicon needed to produce 400 g $SiCl_4$ if the percentage yield is 42.5%.

42.5% = $\frac{400\ g}{\text{theoretical yield}}\times100$

Theoretical yield (mass of $SiCl_4$) = 941.18 g

Mole of $SiCl_4$ = $\frac{941.18\ g}{170.1\ g\ mol}$ **= 5.533 mol**

1 mol $SiCl_4$ ≡ 1 mol Si

5.533 mol $SiCl_4$ ≡ 5.533 mol Si

∴Mass of Si = (5.533 mol)(28.1 g/mol) = <u>155.5 g</u>

ii. If 15 mol of chlorine is used, determine the amount (mole) of unreacted chlorine.

1 mol $SiCl_4$ ≡ 2 mol Cl_2

5.533 mol $SiCl_4$ ≡ 11.066 mol Cl_2

∴ unreacted Cl_2 = (15 - 11.066) mol = <u>3.934 mol</u>

2. a) The wavelength that produces a line, B in Brackett series is 2165.6 nm.

i. Determine the transition that forms the B line.

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), n_1 < n_2$$

$$\frac{1}{2.1656\times10^{-6}m} = 1.097\times10^7 m^{-1}\left(\frac{1}{4^2} - \frac{1}{n_2^2}\right)$$

$n_1 = 7$

∴<u>n =7 to n=4</u>

ii. Calculate the energy emitted for the transition.

$$\Delta E = R_H\left(\frac{1}{n_i^2} - \frac{1}{n_f^2}\right)$$

$$= 2.18\times10^{-18} J\left(\frac{1}{7^2} - \frac{1}{4^2}\right)$$

= <u>-9.18×10^{-20} J</u>

iii. Another line, C was form with wavelength of 1817.5 nm. Explain qualitatively, whether line B or line C, has higher energy emitted.

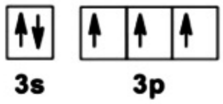
- **Line C has higher energy emitted than line B.**
- **Energy is inversely proportional to wavelength.**
- **Line C has smaller wavelength than line B.**

b) The electronic configuration of element D is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$.

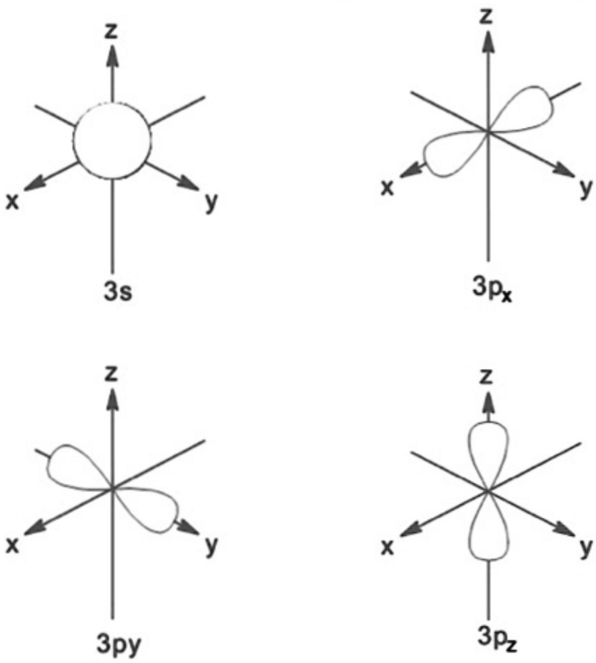
i. Give a set of quantum number for the 9th electron.

(n=2, l=1, m=0, s=+1/2)

ii. Draw the orbital diagram of the valence electrons.



iii. Draw and label the 3D shape of orbitals occupied by the valence electrons.



iv. Explain the filling of the valence electrons in 2(b)(iii) according to the appropriate rule(s)/principle(s).

- **Aufbau principle.**
  Electrons added to the lower energy orbital first. The valence electrons are fill the 3s orbital before 3p orbital. 3s orbital has lower energy than 3p orbital.

- **Hund's rule.**
  When electrons are, add into degenerate orbitals, each orbitals are, filled singly with electron of the same spin before it is pair. 3p orbitals are degenerate orbitals. Therefore, the remaining 3 valence electrons is filled singly in each orbital with the same spin.

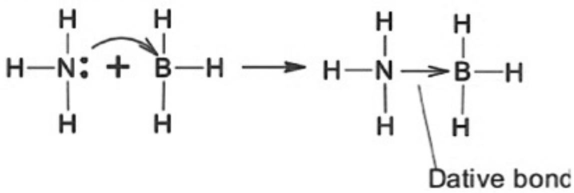
- **Pauli's Exclusion Principle.**
  No two electrons can have same set of quantum number. Only 2 electrons can occupy the same atomic orbitals, thus 2 electrons must be filled in 3s orbital with opposite spin.

3. a) Ammonia, NH_3 and boron triflouride, BF_3 are covalent compounds. NH_3 and BF_3 react to form H_3NBF_3 molecule.

   i. Explain why NH_3 obeys octet rule but BF_3 does not.

- **N in NH_3 is surrounded by 8 valence electrons while B in BH_3 is surrounded by only 6 valence electrons.**
- **BH_3 is incomplete octet.**

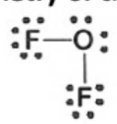
   ii. Show the formation of H_3NBF_3 molecule using Lewis dot symbol and label the bond formed.



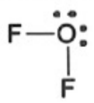
[5 marks]

b) Oxygen difluoride, OF_2 is a strongly oxidizing colourless gas.

   i. Determine the molecular geometry of this molecule.

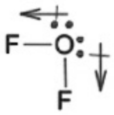


- **Number of electron pair around oxygen atom: 4**
- **Electron pair arrangement: tetrahedral**
- **2 bonding pair and 2 lone pair.**
- **Based on VSEPR theory, the electrons pair will be located as far as possible to minimize the repulsion among them. The lone pair-lone pair repulsion > lone pair-bonding pair repulsion > bonding pair-bonding pair repulsion.**
- **Molecular geometry: V-shape**



ii. Explain whether OF_2 is a polar or non-polar molecule.

- **Polar molecule.**
- **F is more electronegative than O. Therefore, O-F bond is polar.**

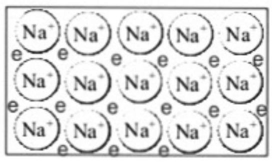


- **Since, molecule OF_2 is unsymmetrical, thus, dipole moment can't cancel each other, $\mu \neq 0$.**

[8 marks]

c) Aluminium and sodium are metals.

i. Explain the formation of metallic bond in sodium using the electron sea model.



- **Metallic bond form in Na metal is the electrostatic forces between positively charge ion, Na^+ and negatively charge free moving electrons.**

ii. Why aluminium has higher boiling point than sodium?

- **Aluminium ion, Al^{3+} is smaller than sodium ion, Na^+. Thus, Al^{3+} has stronger nucleus attraction towards the free moving electrons.**
- **Al has 3 valence electrons and sodium has only 1 valence electron.**
- **Strength of metallic bond is directly proportional to the number of valence electron and inversely proportional to size of ion. Thus, Al has stronger metallic bond than Na.**

4. a) A 10-L cylinder contains 4 g of hydrogen gas and 28 g of nitrogen gas. If the temperature is 31°C,

i. Determine the total pressure of the gas mixture.

Mole of hydrogen gas = 2 mol
Mole of nitrogen gas = 1 mol
Total mol = 3 mol

$$P_T = \frac{n_T RT}{V}$$

$$= \frac{(3\ \text{mol})(0.08206\ \text{atm L mol}^{-1}\ \text{K}^{-1})(304.15\ \text{K})}{(10\ \text{L})}$$

= <u>7.49 atm</u>

ii. Calculate the partial pressure of hydrogen gas.

$$P_{H_2} = X_{H_2} P_T$$

$$= \left(\frac{2}{3}\right)(7.49\ \text{atm})$$

= <u>4.99 atm</u>

iii. What will happen if the gaseous mixture is heated to 550°C?

- **At high temperature, the average kinetic energy of gas particles will increase and cause the reaction between hydrogen and nitrogen to occur producing gas ammonia.**
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b) Under the same condition of temperature and density, determine which gas behaves less ideally: CH_4 or SO_2

molar mass CH_4 = 16 g/mol
molar mass SO_2 = 64.1 g/mol

- **SO_2 will behave less ideally.**
- **SO_2 has higher molecular weight compare to CH_4.**
- **Higher molecular weight will have higher volume of particles. Thus, volume of SO_2 > CH_4.**
- **Strength of intermolecular force of SO_2 > CH_4.**

c) In an experiment when gelatin was added to water, the water became viscous. Explain the relationship between viscosity and intermolecular forces.

- **Viscosity is the resistance of fluid to flow.**
- **Higher intermolecular forces will have higher viscousity.**

5. The equation of simulated photosynthesis reaction is represented by

$$6CO_2(g) \ + \ 6H_2O(g) \ \rightarrow C_6H_{12}O_6(g) \ + \ 6O_2(g) \qquad \Delta H^o = +ve$$

At 31°C, the following equilibrium concentrations were found:
$[H_2O] = 7.91\times10^{-2}$ M, $[CO_2] = 9.30\times10^{-1}$ M, $[O_2] = 2.40\times10^{-3}$ M

a) Calculate the equilibrium constant, K_p for the reaction.

$K_c = \dfrac{[O_2]^6}{[CO_2]^6[H_2O]^6}$

$= \dfrac{(2.40\times10^{-3}M)^6}{(9.30\times10^{-1}M)^6(7.91\times10^{-2}M)^6}$

$= \mathbf{1.21\times10^{-9}}$

$K_c = K_p\,(RT)^{\Delta n}$
$= K_p\,(RT)^{-6}$
$K_p = K_c\,(RT)^{-6}$
$= (1.21\times10^{-9})\,(0.08206\times304.15)^{-6}$
$= \underline{5.02\times10^{-18}}$

b) Determine the initial mass of CO_2 involved in the above reaction.

	$6CO_2$	$6H_2O$	$6O_2$
[ ]i	a	b	0
[ ]c	-6x	-6x	+6x
[ ]e	a-6x	b-6x	6x

$6x = 2.40\times10^{-3}$ M
$x = 4\times10^{-4}$ M

$a - 6x = 9.30\times10^{-1}$ M
$a - 2.40\times10^{-3}$ M $= 9.30\times10^{-1}$ M
$a = 0.9324$ M
$[CO_2]_i = 0.9324$ M

Assume V = 1L

Mole of CO_2 = 0.9324 mol
∴ mass of CO_2 = (0.9324 mol)(44 g/mol)
= **41.03 g**

c) Explain how the equilibrium position would be affected for each of the following changes:

i. Water is added.

- **Equilibrium position will shift forward to consume the added reactant.**

ii. Temperature is increased.

- **Equilibrium position will shift forward to absorb heat added.**

6. a) A sample of 0.214 g of unknown monoprotic weak acid, HA was dissolved in 25.00 mL of water and titrated with 0.1 M NaOH. The acid required 27.40 mL of the base to reach the equivalent point.

i. Determine the molarity of the acid.

$$HA(aq) + NaOH(aq) \rightarrow NaA(aq) + H_2O(l)$$

Mole of NaOH = 2.74×10^{-3} mol

1 mol NaOH ≡ 1 mol HA
2.74×10^{-3} mol NaOH ≡ 2.74×10^{-3} mol HA

Molarity = $\dfrac{2.74\times10^{-3}\,mol}{0.025L}$
= **0.1096 M**

ii. Calculate the pH of the solution if 40 mL NaOH is added to the acid solution.
Mole of NaOH = 4×10^{-3} mol

	HA	NaOH	NaA
n_i	2.74×10^{-3} mol	4×10^{-3} mol	0
n_c	-2.74×10^{-3} mol	-2.74×10^{-3} mol	$+2.74\times10^{-3}$ mol
n_f	0	1.26×10^{-3} mol	2.74×10^{-3} mol

Molarity of NaOH = $\dfrac{1.26\times10^{-3}\,mol}{0.065L}$
= 0.0194 M

$pOH = -\log(0.0194\ M)$

$= 1.71$

$\therefore pH = \underline{\underline{12.29}}$

iii. Explain qualitatively the pH of solution at equivalent point.

- **At equivalent point, all acid is completely reacted with base.**
- **The only remaining particles in the solution is salt, NaA solution.**
- **NaA salt is from weak acid and strong base, the anion A^- undergoes hydrolysis forms OH^-.**

  $$A^-(aq) + H_2O \rightarrow HA\ (aq) + OH^-(aq)$$

- **pH > 7.**

b) Magnesium arsenate, $Mg_3(AsO_4)_2$, is used as an insecticides and it is toxic to humans. Calculate the solubility of $Mg_3(AsO_4)_2$ in water at 25°C.
[Given: K_{sp} $Mg_3(AsO_4)_2 = 2.2\times10^{-20}$]

$$Mg_3(AsO_4)_2\ (s) \rightarrow 3Mg^{2+}(aq)\ +2AsO^{3-}(aq)$$

[ ]$_i$	**-**	**0**	**0**
[ ]c	**-**	**+3x**	**+2x**
[ ]e	**-**	**3x**	**2x**

$K_{sp} = [Mg^{2+}]^3[AsO_4^{3-}]^2$

$2.2\times10^{-20} = (3x)^3(2x)^2$

$= 108x^5$

$x = 4.59\times10^{-5}$

$\therefore$Solubility = $\underline{\underline{4.59\times10^{-5}\ M}}$